THE TOTAL DERIVED FUNCTOR OF A COMPOSITION OF MORPHISMS OF RINGED SPACES IS THE COMPOSITION OF THE TOTAL DERIVED FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This card was re-admitted after a gate audit: it had been excluded only because the chapter's retained preamble lacks the slogan environment. The statement carries the author's own slogan, which the author's preamble comments out; the card carries a per-card preamble copy formed from the chapter retained preamble plus the project's own three display environments copied verbatim from sources/stacks-chow/preamble.tex lines 7-12.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The label lemma-higher-direct-images-compose is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Composition criterion for total derived functors. Derived Categories, Lemma derived-lemma-compose-derived-functors, invoked at proof lines 2284-2285. Supplies the reduction to the acyclicity condition.
- Pushforward of sheaves is compatible with composition. Sheaves, Lemma sheaves-lemma-pushforward-composition, invoked at proof lines 2286-2287. Supplies that the underlying functors compose.
- Pushforward of an injective module is acyclic, and the description of higher direct images. Same chapter, Lemmas lemma-pushforward-injective and lemma-describe-higher-direct-images, invoked at proof lines 2288-2292. Supplies the acyclicity verification.

Source reference. Original lemma 015M: sources/stacks-chow/tags.txt, source lines 1513–1513. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01E4: sources/stacks-cohomology/pinned-original-source.tex, source lines 592–592. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EX: sources/stacks-cohomology/pinned-original-source.tex, source lines 1779–1779. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 008E: sources/stacks-chow/tags.txt, source lines 314–314. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of ringed spaces. In this case $Rg_* \circ Rf_* = R(g \circ f)_*$ as functors from $D^+(X) \to D^+(Z)$.*

Proof. We are going to apply Derived Categories, Lemma 015M. It is clear that $g_* \circ f_* = (g \circ f)_*$, see Sheaves, Lemma 008E. It remains to show that $f_*\mathcal{I}$ is g_*-acyclic. This follows from Lemma 01EX and the description of the higher direct images $R^i g_*$ in Lemma 01E4. □